PROJECT REPORT

Project Title: Intelligent Spam Email Detector using Machine Learning

Course: Artificial Intelligence Lab

Date: 09, December 2025

Name	Student ID	Role
Junayet Mithu	0242310005341400	Data Analyst & Model Trainer
Rabby Khan	0242310005341478	Backend Developer & Logic Integration
Fardin Hasan	0242310005341393	Frontend Designer & UI/UX

1. Abstract

Email spam is a major cybersecurity threat, leading to phishing attacks, financial fraud, and malware distribution. Traditional spam filters often fail to detect modern, sophisticated attacks. This project presents an **Intelligent Spam Detector** that uses Machine Learning (Naive Bayes) to classify emails with **97% accuracy**. Unlike standard filters, our system includes **Explainable AI (XAI)** features that analyze the psychological triggers behind spam (Fear, Greed, Urgency) and calculate a real-time risk score, providing users with a safer and more educational email experience.

2. Problem Statement

Millions of spam emails are sent daily. While many are annoying advertisements, a significant portion attempts to steal data or money.

- **The Problem:** Most users cannot distinguish between a legitimate "Urgent" email from their boss and a fake "Urgent" email from a hacker.
- **The Solution:** We need an automated system that not only blocks spam but explains *why* a message is dangerous, helping users improve their own security awareness.

3. Unique Features (Novelty)

Our system goes beyond simple "Spam vs. Not Spam" classification. We implemented three unique features:

1. Psychological Trigger Detector:

   The system analyzes the text to identify the social engineering tactic used by the attacker. It flags if the email is using FEAR (e.g., "Arrest warrant"), GREED (e.g., "Win money"), or URGENCY (e.g., "Act now").

2. Probabilistic Risk Score:

   Instead of a rigid Yes/No result, the system provides a confidence percentage (e.g., "98.5% Risk of Spam"), allowing users to judge borderline cases.

3. Automated Counter-Measure (Auto-Reply):

   As an active defense mechanism, the system generates a humorous, automated response to confirmed spam emails, designed to waste the scammer’s time and resources.

4. Methodology

4.1 Dataset

We utilized the **SMS Spam Collection Dataset** from Kaggle, containing **5,572 labeled messages**.

- **Spam Messages:** 747
- **Legitimate (Ham) Messages:** 4,825
- **Preprocessing:** We cleaned the data by removing special characters and converting all text to lowercase.

4.2 Algorithm Used

- **TF-IDF Vectorization:** We used *Term Frequency-Inverse Document Frequency* to convert text data into numerical vectors. This highlights unique words (like "lottery") while ignoring common words (like "the").
- **Multinomial Naive Bayes:** This probabilistic algorithm was chosen because it is highly effective for text classification and requires less training time than Neural Networks while maintaining high accuracy.

5. Implementation Details

5.1 Tools & Technologies

- **Language:** Python 3.9
- **Backend Framework:** Flask (Lightweight web server)
- **Machine Learning Library:** Scikit-learn (sklearn)
- **Data Handling:** Pandas
- **Frontend:** HTML5, CSS3

5.2 System Architecture

1. **Input:** User pastes email content into the web interface.
2. **Vectorization:** The backend loads `vectorizer.pkl` to translate text into numbers.
3. **Prediction:** The `spam_model.pkl` calculates the probability of the input being spam.
4. **Feature Logic:** The system scans for keywords (Psychological Triggers) and generates an Auto-Reply if needed.
5. **Output:** The result is displayed on the user's screen.

6. Results and Performance

After training our model on the dataset, we achieved the following metrics:

- **Accuracy: 96.86%**
- **Precision (Spam): 100%** (We successfully eliminated False Positives).
- **Recall (Spam): 77%**

The high precision ensures that our system is safe for business use, as it does not accidentally block important legitimate emails.

7. Conclusion

This project successfully demonstrates how Machine Learning can solve real-world cybersecurity problems. By combining a high-accuracy Naive Bayes model with user-friendly features like the **Psychological Trigger Detector**, we have created a tool that protects users while educating them about common scam tactics.

8. Future Scope

- **Deep Learning Integration:** Implementing LSTM (Long Short-Term Memory) networks to better understand context in very long emails.
- **Browser Extension:** Converting the web app into a Chrome extension for real-time protection inside Gmail.
- **Multi-Language Support:** expanding the dataset to detect spam in Bengali and other languages.

Github Link: https://github.com/rabbykhanswe/AI-Project